

Managing Exchange Rate Exposure with Hedging Activities: New Approach and Evidence from Korea

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Abstract

We examine the effectiveness of firms' operating and financial activities in managing their exchange rate exposures. Unlike previous studies, we measure the *expected* exchange rate exposure which reflects exchange rate risk associated with firms' inherent business activities prior to the usage of hedging activities and compare it with the *observed* exchange rate exposure which reflects the effects of firms' hedging activities. We show that the effectiveness of hedging activities depends on the underlying characteristics (e.g., direction) of firms' inherent expected exchange rate exposures. While firms with positive expected exposures reduce their exposures through currency derivatives, internal transactions with foreign subsidiaries, and the issuance of foreign currency debt, firms with negative expected exposures do so only through exchange rate pass-through activities. Our results strongly suggest that in order to uncover the effectiveness of firms' hedging activities, one must consider both the conditions in the firms' product market (e.g., export and import ratios and profit margin) and the direction of exchange rate exposure.

JEL Classification: F31; G15

Key words: Hedging activities; Expected exchange rate exposure; Observed exchange rate exposure; Korean firms

1. Introduction

A large body of studies has examined the measurement of exchange rate exposure and its effect on firm value with inconclusive evidence on the existence of exchange rate exposure. Adler and Dumas (1984) show that a firm's exchange rate exposure can be measured from a regression model of the firm's exposed asset prices expressed in the domestic currency against the foreign currency. Jorion (1990, 1991) measures the exchange rate exposure using stock returns but fails to show evidence that exchange rate risk is an important determinant of firm value. Subsequent studies have estimated exchange rate exposures using regression models of stock returns and exchange rate changes. While a few of these studies report relatively significant levels of exchange rate exposure (e.g., He and Ng, 1998, Dominguez and Tesar, 2006) and exchange rate risk premium (e.g., Dumas and Solnik, 1995; Kwon, Bae, and Chung, 2005), the majority of these studies offer evidence that is not in supportive of the existence of exchange rate exposure or the need to manage exchange rate risk.

As possible explanations for the lack of strong evidence on exchange rate exposure, earlier studies offered several factors including the measurement problems associated with the estimation of exchange rate exposure and the delayed effects of exchange rate changes (see, e.g., Bartov and Bodnar, 1994). Bartram and Bodnar (2007) demonstrate, however, that correcting and considering these factors still do not lead to an increase in the significance levels of the estimated regression coefficients of exchange rate exposure. Instead, they note that the evidence of insignificant regression coefficients of exchange rate exposure reported in previous studies is not surprising at all. They argue that it is not due to the estimation problem but mainly due to firms' management of exchange rate exposure. Bartram, Brown, and Minton (2010) offer evidence supporting the Bartram and Bodnar's view by reporting that firms indeed manage a large portion of their exchange rate exposures.

In this paper, we examine two main issues pertaining to exchange rate exposure—the measurement of exchange rate exposure and the effectiveness of firms' managing activities of exchange rate exposure. Our paper takes a very different approach to the examination of these issues from existing studies, which

have focused on firms' observed stock returns. As Bodnar and Martson (2002) and Bartram, Brown, and Minton (2010) point out, however, an effort to measure a firm's exchange rate exposure using the firm's observed stock returns would fail to uncover the existence and the true level of the firm's exchange rate exposure. Because a firm's observed stock returns would already have reflected the outcomes of the firm's hedging activities in managing exchange rate exposure, an examination of the observed stock returns is more likely to reveal insignificant exchange rate exposure for the firm.

In order to overcome this measurement problem, we separately measure *expected* exchange rate exposure, which reflects firms' exposure levels related to their basic business activities prior to their usage of any hedging activity, and *observed* exchange rate exposure, which reflects the effects of the firms' other activities for hedging as well. We then compare the observed exchange rate exposure with the expected exchange rate exposure and analyze their difference. Our logic is that because this difference captures the outcome of firms' hedging activities, its analysis will reveal the true effectiveness of firms' activities to manage exchange rate exposure by controlling external factors that might affect the firms' exchange rate exposures. That is, if the observed exchange rate exposure is excluded from the expected exchange rate exposure, one can isolate the exchange rate exposure associated with firms' hedging activities. Hence, the significantly large expected exchange rate exposure relative to the observed exchange rate exposure would indicate a substantial reduction in firms' exchange rate exposures and thus the effectiveness of a hedging activity.

The main contribution of our paper is to empirically measure expected exchange rate exposure for a comparison to observed exchange rate exposure by explicitly considering the direction of exchange rate exposure. Bodnar and Marston (2002) develop a theoretical model of the expected exchange rate exposure using a firm's export and import ratios as its key determinants but offer no empirical evidence. Bartram, Brown, and Minton (2010) empirically measure the expected exposure, but their analysis is limited to automobile companies, and more critically, fails to consider the direction of the expected exchange rate exposure. The difficulty of measuring the expected exchange rate exposure rests primarily on the notion

that data on firms' import ratios are regarded as the firms' trade secrets and thus are not publicly available. In this paper, we proxy firms' import ratios by the input shares of imported goods relative to the sales of the sector where firms belong to.

Firms' management of exchange rate exposure generally involves both operating and financial activities. The operating activities include domestic-currency invoicing, matching and offsetting, and exchange rate pass-through, among others. In particular, several studies offer evidence supporting that firms pass through their exchange rate exposures to sales prices with and without the adjustment of mark-ups, especially in the face of decreasing local currency value (Allayannis and Ihrig, 2001; Campa and Goldberg, 1999; Gagnon and Knetter, 1995; Knetter, 1989, 1993, 1994; Krugman, 1987; Manon, 1995).¹ The financial activities for hedging include the use of currency derivatives, financings through foreign-currency denominated debt, and internal transactions with foreign subsidiaries, among others. Existing studies offer inconclusive evidence on the effectiveness of the foreign currency denominated debt in managing exchange rate exposure (see, e.g., Bae and Kwon, 2013; Bartram, Brown, and Minton, 2010; Clark and Judge, 2009). Similarly, the existing literature shows mixed evidence on the effectiveness of currency derivatives. While several studies document positive hedging effects of financial derivatives (e.g., Allayannis and Weston, 2001; Allayannis and Ofek, 2001; Bartram, Brown, and Conrad, 2011; Bartram, Brown, and Minton, 2010; Clark and Judge, 2009; Graham and Rogers, 2002; Guay, 1999), a good number of other studies cast doubt on their effectiveness in exposure management (e.g., Bali, Hume, and Martell, 2007; Guay and Kothari, 2003; Hentschel and Kothari, 2001).

In this paper, we examine both operating and financial hedging activities for exposure management and argue that the mixed evidence on the effectiveness of hedging activities is largely attributed to the

¹ Through a survey of large euro-area companies, Döhring (2008) shows that many firms make use of operational hedges including matching foreign currency revenues with expenditures in the same currency and the international diversification of sourcing, production, and sales.

failure to consider the underlying characteristics of exchange rate exposure for each individual firm.² Because exchange rate exposure can be either positive or negative, which would require different hedging activities, it is important to identify the correct directions of a firm's exchange rate exposure for the effectiveness of hedging activities.

We focus on manufacturing companies in Korea, one of the premier developing countries, for empirical evidence. Korean manufacturing firms have long depended on international trades and foreign capital with steady increases in their foreign business operations and direct investments over the last decade or so, which have made their firm values highly sensitive to exchange rate changes.³ In addition, the current accounting system that requires firms to report in the concurrent year's balance sheet the gains and losses in asset values associated with exchange rate changes has also contributed to the exchange rate exposure of Korean firms. Consequently, with the larger swings in the exchange rates recently, Korean firms have had much greater needs to manage their exchange rate risk than ever. It is well documented that Korean firms employ various hedging tools including exchange rate pass-through, internal transactions with foreign subsidiaries, and derivatives products, among others (see, e.g., Jung and Kwon, 2007). In this regard, Korean manufacturing firms are ideal research targets for the examination of the exchange rate exposure and the effectiveness of various hedging activities for exposure management.

The results of our paper support the notion that the effectiveness of hedging activities depends on the underlying characteristics (e.g., direction) of firms' inherent exchange rate exposures, that is, the expected exchange rate exposures in our paper. To be more specific, firms with positive expected exchange rate exposures are shown to reduce their exposures through selling currency forwards/futures, internal transactions with foreign subsidiaries, and the issuance of foreign currency denominated debt, whereas firms with negative expected exchange rate exposures do so only through exchange rate pass-

² Other factors that may contribute to the mixed evidence include differences in sample firms, time periods, and/or measurement of model specifications.

³ Since the Asian financial crisis in late 1997, Korean firms have been exposed to an unprecedented level of foreign exchange risk primarily due to the adoption of the flexible exchange rate system in late 1990s.

through associated with exchange rate changes. In addition, our analysis of firms with significant observed exchange rate exposures offers evidence that structured currency forwards deteriorate firms' exchange rate exposures. Our results also show that while some firms tend to over-manage exchange rate exposures, the majority of firms properly use hedging activities to manage exchange rate exposure.

The overall results of our paper strongly suggest that for the effective management of exchange rate exposures firms consider both the pre-conditions in the product market (e.g., export and import ratios and profit margin) that they face before engaging in hedging activities and the underlying characteristics (e.g., direction) of exchange rate exposure. In particular, the effectiveness of hedging activities can vary significantly depending on the degree of responses in their product markets to exchange rate changes. Hence, relying simply on the regression results of observed stock returns without considering these notions would likely yield very limited and unreliable empirical evidence.

Our paper is organized as follows. Section 2 presents the research design and data including methods to measure the expected exchange rate exposure and the observed exchange rate exposure. Section 3 reports the empirical results, with the summary and conclusions in Section 4.

2. Research design and data

2.1. Measurement of expected exchange rate exposure

We measure the expected exchange rate exposure inherent in the firm's basic business activities by the estimation models used by Bodnar and Marston (2002). A firm's value can be expressed using future cash flows as follows:

$$V = \sum_{t=1}^{\infty} \frac{CF_t}{(1+\rho)^t} \quad (1)$$

where V = firm value; CF_t = expected future cash flows, measured as the difference between income after taxes and net investment); ρ = discount rate. Assuming constant cash flows and zero net investment for simplicity, the firm value will be represented by:

$$V = \frac{CF}{\rho} = \frac{(1-\tau)}{\rho} \pi \quad (2)$$

where τ = corporate tax rate; π = profit before taxes.

If we denote the exchange rate of local currency relative to foreign currency as Er , then a change in firm value relative to a change in the exchange rate, that is, exchange rate exposure, can be expressed as dV/dEr . With a constant corporate tax rate of τ and a constant discount rate of ρ , the exchange rate exposure can be expressed as:

$$\frac{dV}{dEr} = [(1-\tau)/\rho] \frac{d\pi}{dEr} \quad (3)$$

Changes in exchange rates typically affect firms engaging in the following types of business activities: (1) producing domestically and exporting to foreign countries; (2) producing or selling final products using imported raw materials; and (3) producing same or different products in foreign subsidiaries. As a way to consider all these types of business activities, we take the case of a multinational enterprise that produces products made of domestic or foreign raw materials and sells them both domestically and in foreign countries. The net profit, π , of such a firm is given by:

$$\begin{aligned} \pi = & XD(X) + ErX^*D^*(X^*) - C(X + X^*) - ErC^*(X + X^*) \\ & + ZF(Z) + ErZ^*F^*(Z^*) - K(Z + Z^*) - ErK^*(Z + Z^*) \end{aligned} \quad (4)$$

where X, X^* = sales of domestically-produced products domestically and abroad, respectively; $D(X), D^*(X^*)$ = prices of domestically-produced products sold domestically and abroad, respectively; Z, Z^* = sales of foreign subsidiary-produced products domestically and abroad, respectively; $F(Z), F^*(Z^*)$ = prices of foreign subsidiary-produced products sold domestically and abroad, respectively; C, C^* = expenses of domestically-produced products in local currency and in foreign currency, respectively; and K, K^* = expenses of foreign subsidiary-produced products in local currency and in foreign currency, respectively.

We denote R as total sales measured by $X D(X) + Er X^* D^*(X^*) + Z F(Z) + Er Z^* F^*(Z^*)$; M as total expenses measured by $C(X+X^*) + Er C^*(X+X^*) + K(Z+Z^*) + Er K^*(Z+Z^*)$; h_l as the ratio of foreign

sales to total sales measured by $Er [X^*D^*(X^*) + Z^*F^*(Z^*)]/R$; h_2 as the ratio of foreign expenses (or expenses in foreign currency) to total expenses measured by $Er [C^*(X+X^*)+K^*(Z+Z^*)]/M$, and r as profit margin measured by π/R . Then, if the exchange rate exposure is expressed in the form of elasticity, the elasticity of a firm value in terms of net profit to a change in the exchange rate, δ , can be presented by:⁴

$$\delta = \frac{d \ln \pi}{d \ln Er} = h_1(R/\pi) - h_2(M/\pi) = (h_1/r) - h_2((1/r) - 1) = h_1 + (h_1 - h_2)(\frac{1}{r} - 1) \quad (5)$$

Equation (5) shows that the expected exchange rate exposure inherent in the firm's basic business activities is determined by the firm's foreign sales or export ratio (h_1), foreign expense (that is, expense of imported materials for final products) or import ratio (h_2), and profit margin (r). The relationship between export ratio and import ratio is a key factor in determining both the existence and the magnitude of a firm's exchange rate exposure, and the firm's profit margin plays a role in the determination of the magnitude of exchange rate exposure. In equation (5), if a firm's export ratio is greater than its import ratio, the firm will be exposed to greater exposure where the firm's value increases (decreases) as the exchange rate goes up (down). It is also shown that the expected exchange rate exposure becomes bigger as the firm's ability to generate profits, or the profit margin, is lower.

Because data on a firm's import ratios are regarded as the firm's trade secrets and thus are not publicly available, we proxy a firm's import ratio by relating the firm's sales composition to the imported input share of sales of the sector or industry where the produced goods belong. The imported input shares of sector sales are collected from the input-output tables reported by the Bank of Korea.⁵ For example, if a firm produces goods belonging to the manufacturer of pulp, paper, and paperboard (KSIC 17), we use the corresponding sector's imported input share of 25.56% (2007 year basis) as a proxy for the firm's import ratio. If a firm is diversified and thus produces multiple goods, we compute and employ the weighted average of the imported input shares of sector sales for the multiple goods as the firm's import ratio. For

⁴ For the detailed derivation of equation (5), see Bodnar and Marston (2002).

⁵ The input-output tables are widely used by economists as a basis to determine whether goods are capital-intensive or labor-intensive.

instance, a firm's sales consist of \$30 million in sector A and \$20 million in sector B. Then the shares of sectors A and B of total sales are 60% and 40%, respectively. If the imported input shares of sector sales in the sectors A and B are 0.1 and 0.2, respectively, as reported in the Input-Output Table, then the firm's imported ratio is proxied as 14% (= 60% * 0.1 + 40% * 0.2).

2.2. *Measurement of observed exchange rate exposure*

We measure the observed exchange rate exposure present in the firm's stock returns by employing the following market model (Jorion, 1990):

$$Sr_{i,t} = \alpha_{0,i} + \alpha_{1,i}Mr_t + \alpha_{2,i}Err_t + \varepsilon_{i,t} \quad (6)$$

where Sr = a firm's daily stock returns; Mr = KOSPI daily stock market returns; Err = changes in daily exchange rate (Er); and all variables are measured based on daily closing prices and exchange rate.⁶ In equation (6), the exchange rate is the local currency of Korean won (KRW) relative to the foreign currency of U.S. dollar (USD), and the estimated model coefficient of $\alpha_{2,i}$ captures the observed exchange rate exposure of firm i . It is worth noting that as $\alpha_{2,i}$ measures residual exchange rate exposure after sorting out the market-related exposure of a firm's stock returns, a zero coefficient of $\alpha_{2,i}$ implies that a firm has the same exposure as the market portfolio, not that the firm is unaffected by exchange rate changes (Bodnar and Wong, 2003).

2.3. *Analysis of difference in expected and observed exchange rate exposures*

The observed exchange rate exposure measured from the relation between a firm's stock returns and changes in the exchange rate represents the combined outcome of the expected exchange rate exposure that results from the firm's basic business activities and the effects of other activities (e.g., using foreign

⁶ We employ daily stock returns for firms and the value-weighted KOSPI market index in equation (6). Due to the potential impacts of different market index and return horizon, we perform robustness tests with alternative variables and report their results in the later part of our paper.

currency denominated debt and currency derivatives) on exchange rate exposure. Therefore, it would be impossible to analyze separately what level of exchange rate exposure is inherent in the firm's basic business activities and whether the firm's other activities to manage exchange rate exposure are effective by examining the observed exchange rate exposure alone. In order to correctly measure the effects of the firm's managing activities of exchange rate exposure, it is necessary to first estimate the expected exchange rate exposure inherent in the firm's basic business activities separately. Then, if the observed exchange rate exposure is excluded from the expected exchange rate exposure, one can isolate the exchange rate exposure associated with the firm's managing activities of exchange rate exposure. Hence, if one analyzes the difference between the expected exchange rate exposure and the observed exchange rate exposure, called exchange rate exposure difference (EDIF), one should be able to find the relationship between managing activities of exchange rate exposure and the firm's exchange rate exposure. Because EDIF reflects the outcome of the firm's managing activities of exchange rate exposure, an analysis of EDIF by controlling external factors that might affect the firm's exchange rate exposure will reveal the effectiveness of the types of activities firms use to manage exchange rate exposure. In Figure 1, the two types of exchange rate exposure are presented in relation to the firms' hedging activities.

Firms with insignificant observed exchange rate exposure are likely to have managed effectively their exchange rate exposure by considering external conditions (Bartram and Bodnar, 2007; Bartram, Brown and Minton, 2010). Hence, if one measures EDIF of firms with insignificant observed exchange rate exposure and estimates regression models of EDIF as dependent variable and firms' managing activities of exchange rate exposure as explanatory variables, one would be able to find the effectiveness of the types of activities used for managing exchange rate exposure. Based on this discussion, we estimate the following regression model (for simplicity we omit the subscript t for year):

$$\begin{aligned}
 EDIF_i = & b_0 + b_1FWD_i + b_2NSFWD_i + b_3NFCFIN_i + b_4PASS_i + b_5INTTR_i + b_6DIVER_i \\
 & + b_7FSIZE_i + \sum_{j=1}^J b_{7+j}IDUMMY_{j,i} + \sum_{y=1}^Y b_{7+J+y}YDUMMY_{y,i} + \eta_i
 \end{aligned} \tag{7}$$

where *EDIF* = exchange rate exposure difference, measured by the difference between expected and observed exchange rate exposure. In regression model (7), *FWD*, *NSFWD*, *NFCFIN*, *PASS*, *INTTR* are testing variables, representing firms' hedging activities for exposure management, and the other variables are used as control variables. A brief description of explanatory variables along with their measurements is also given below:

Testing variables of hedging activities

FWD represents the transaction (buy and sell) amount of currency derivatives including currency forwards, currency futures, and currency risk insurance.⁷ We construct *FWD* by separating buy transactions (equivalent to long positions) and sell transactions (equivalent to short positions), yielding two variables of *FWD-Buy* and *FWD-Sell*.

NSFWD represents the net position in structured forward contracts, measured by the difference between short and long position in structured forward contracts. The short (long) position in structured forward contracts combines buying (selling) put options and selling (buying) call options, making a position identical to selling (buying) currency forward. A currency KIKO (knock-in knock-out) option is an example of the structured forward contract and was one of the instruments widely used for hedging exchange rate (in particular, US dollars) risks in the Korean financial markets until late 2008.⁸

NFCFIN represents net amount of foreign currency financing, measured by the difference between total foreign currency financing and swap transaction amount. Total foreign currency financing is the sum

⁷ Currency risk insurance is a part of export insurance system for exporting and importing firms offered by Korea Trade Assurance Corporation (KTAC) since 2000 and works in a similar way to the currency forward contract offered by financial institutions. It recovers a firm's losses and collects the firm's profits associated with exchange rate changes by comparing the exchange rate guaranteed by KTAC and the actual exchange rate at the settlement time.

⁸ The KIKO option is designed to offer positive payoffs to the option holder when KRW moderately appreciates up to a certain predetermined rate against USD; in exchange, the option holder is obligated to take negative payoffs when the KRW value depreciates significantly (see Khil and Suh, 2010). As the KRW depreciated unexpectedly during the global financial crisis in 2007 and 2008, however, the KIKO option incurred substantial losses to the option holders, known as KIKO disaster. According to the Korean Financial Services Commission, as of June 2008, 519 firms held the KIKO options in the outstanding amount of USD10.1 billion, and 68 firms holding overhedged KIKO positions (amounts of KIKO options exceeding their export amounts) reported financial losses of USD384 million, which far exceeded financial gains of USD142 million from their USD export revenues.

of foreign-currency denominated short-term debt and long-term debt, liquidity long-term debt, and foreign currency denominated bonds. Swap transaction amount includes currency swaps and currency interest swaps denominated in foreign currencies.

PASS represents a firm's exchange rate pass-through ratio. Bodnar, Dumas and Marston (2002) and Allayannis and Ihrig (2001) show that a firm's path-through is an important determinant of exchange rate exposure along with the competitiveness structure of the firm's industry. Because data on a firm's exchange rate pass-through ratio are not publicly available, it is practically impossible to measure this ratio. In this paper, we estimate each firm's pass-through ratio using the pass-through ratios of sales items for the industry to which the firm belongs in response to the exchange rate changes as reported by the Bank of Korea.⁹ As an example, let's assume that a firm produces two products, A and B with sales of \$10 million and \$20 million, respectively. If the pass-through ratios of sales items A and B in the industry are 50% and 20%, respectively, (as reported by the Bank of Korea based on KSIC industry classifications), then the firm's pass-through ratio is computed as 30% ($=10/30 \times 50\% + 20/30 \times 20\%$).

INTTR represents the degree of internal transactions of each firm with foreign subsidiaries, measured by the amount of internal transactions relative to the firm's total sales.

Control variables

DIVER represents the degree to which a firm's operations are diversified into other lines of business. *DIVER* is included in the regression model to consider the effect of the firm's diversification on the relation between a firm's derivatives transactions and its risk level reported in previous studies (e.g., Bartram, Brown, and Conrad, 2011). For *DIVER*, we employ the widely-used Caves weighted index of diversification based on the firm's sales (Caves et al., 1980). A higher value of *DIVER* indicates a greater diversification of a firm's operations.

FSIZE represents firm size, and is measured by the natural log of the sum of the market values of

⁹ This measurement approach is similar to the one for the measurement of a firm's import ratio.

common stock and preferred stock and the book value of debt.

IDUMMY is industry dummies, spanning twenty-two industries from food and beverage (KSIC 10) to publishing (KSIC 58). *IDUMMY* is included to consider the potential differences in industries with respect to the risk level and the ease of hedging (Jin and Jorion, 2006). The industry classification is based on two-digit KSIC. Since firms may change their industry classifications as time goes by, the sales item with the highest actual sales is used to identify the firm's major industry.

YDUMMY is year dummies and included to control for fixed-time effects such as sudden rebounds and adjustments in exchange rates during the sample period.

The variables of *FWD* (*FWD-Buy* and *FWD-Sell*), *NSFWD*, and *NFCFIN* are measured as relative to firm size, whereas *PASS*, *INTTR*, and *DIVER* are measured as relative to the firm's sales.

Collectively for the five testing variables of firms' hedging activities, we offer a summary of the interpretation of EDIF with respect to the effectiveness of hedging activities given different signs and magnitudes of the observed and expected exchange rate exposure measures in Table 1.

2.4. Data

The sample firms in our paper consist of all Korean industrial firms except for financial firms listed on the Korea Exchange for the period of 2007-2009. 2007 year was when the exchange rates of KRW against USD steadily decreased (i.e., the value of KRW increased). In contrast, the exchange rates of KRW against USD increased rapidly due to the global financial crisis throughout the second half of 2008, continued to increase in early 2009, and then declined in late 2009.

The data for export ratios and operating margin ratios necessary to estimate the expected exchange rate exposure are collected from TS2000, the database of Korea Listed Companies Association. The data on stock returns necessary to estimate the observed exchange rate exposure are collected from the KCMI-SD database of Korean Capital Market Institute. We estimate the observed exchange rate exposure on the annual basis using daily stock returns. For exchange rate changes, we use changes in nominal exchange

rates, rather than changes in real exchange rates, considering the relatively smaller changes in daily inflation rates.

For each firm's derivatives transactions, we collect related information by searching footnotes of firms' business reports and audit reports available on the Electronic Disclosure System of Korea Financial Supervisory Service (<http://dart.fss.or.kr/>). Regarding the firm's exchange rate pass-through ratio, we rely on the firm's sales reports and the industry report on pass-through ratios classified by sales items as reported by the Bank of Korea.

3. Empirical results

3.1. Measures of expected exchange rate exposure and observed exchange rate exposure

Tables 2 and 3 report mean values of expected and observed exchange rate exposures, estimated from equations (5) and (6), respectively, and other variables of interest by industry¹⁰ for two groups of firms—firms with statistically insignificant and significant (at the 10% level) observed exchange rate exposure. For each firm, the exposure is measured on an annual basis and then pooled together to produce an average value over the three-year period. Equation (5) used to estimate the expected exchange rate exposure assumes a firm's maximization of its operating profits. Hence, it is reasonably expected that for low-performing firms in a given year, the estimated expected exchange rate exposure would have little practical meaning. Accordingly, we exclude firms whose profit margins in a given year are less than 25% of the average of all sample firms in the same year.¹¹ In both tables, we include firms whose firm-year observations are less than six into the industry classification of 'Other'. Panels A and B of both tables

¹⁰ See Appendix for the Korea Standard Industrial Classification (KSIC) codes and industry definitions of firms used in our study.

¹¹ Nance, Smith and Smithson (1993) show that U.S. firms hedge mainly to reduce expected tax liability, volatility of taxable income, and thus expected costs of financial distress. Their findings suggest that low-performing firms also have reason to hedge to reduce expected costs of financial distress as these firms have even more at stake when unhedged and suffer currency losses. In order to examine the effect of low performance on firms' hedging activities, we perform robustness tests later with different thresholds of profit margin.

show results for firms with positive and negative expected exchange rate exposure, respectively.

Firms in Panel A of Table 2 are characterized by the relationship that their values increase (decrease) when the exchange rate goes up (down) based on the expected exchange rate exposure. Looking at the currency derivatives positions (*FWD-Buy* and *FWD-Sell*), sell positions are larger by about five times than buy positions (0.057 vs. 0.011) in all industries except for the three industries of KSIC 17 (pulp, paper, and paper goods manufacturers), KSIC 21 (medicine and medical supplies manufacturers) that engage in buying currency derivatives only, and KSIC 46 (wholesalers and merchandise brokers) that has the same ratio of buy and sell positions of currency derivatives. The largest sell transactions of currency derivatives are done by KSIC 31 (other transportation equipment manufacturers) with its transactions of sell positions being about 9 times those of buy positions (0.884 vs. 0.101).

The foreign currency financing amount is on average about 3.6% of firm size in all industries, with its highest ratio of 12.1% in KSIC 13 (textile manufacturers). The magnitude of currency swaps and currency interest swaps for the sample firms is about 37% ($= 0.013/0.036$) of foreign currency financing amount. KSIC 17 shows the largest net foreign currency debt of 11.6%, while KSIC 31 has the largest net foreign currency assets of 10.2%, relative to firm size.

When industries are ranked by the magnitude of expected exchange rate exposure, KSIC 46 captures the top place, followed by KSIC 13, KSIC 31, and so on. Firms in KSIC 46 engage in the largest transaction amount of buying currency derivatives (relative to firm size) and foreign currency financing and possess net foreign currency debt. On the contrary, this industry has relatively low transaction amount of selling currency derivatives and lower swap transaction amount. The observed exchange rate exposure has the same positive sign as the expected exchange rate exposure, suggesting little over-management of exchange rate exposure.

Firms in KSIC 13 possess substantially greater sell transaction amount of currency derivatives than their buy transaction amount, and relatively large amount of foreign currency financing, satisfying the condition to reduce the positive exchange rate exposure. On the other hand, the possession of relatively

larger foreign currency assets weakens these firms' ability to reduce the positive exchange rate exposure. Combined together, these activities lead to negative observed exchange rate exposure for firms in KSIC 13, implying an over-management of exposure.

For firms in KSIC 31, the sell transaction amount of currency derivatives is almost 9 times their buy transaction amount, and the swap transaction amount relative to foreign currency financing is low, leading to a favorable condition to reduce the expected exchange rate exposure. On the other hand, this industry holds relatively large foreign currency assets, which weaken the reduction of exchange rate exposure. Combined with the negative observed exchange rate exposure, the findings suggest an over-management of exchange rate exposure for firms in KSIC 31.

Panel B shows results for firms that are characterized by decreases (increases) in their firm values when the exchange rate goes up (down). Overall, these firms have substantially larger transaction amount of buying currency derivatives than selling currency derivatives (25 times), engage swap transactions equal to about 47% of foreign currency financing amount, and possess net foreign currency debt. Among industries, KSIC 35 (electricity, gas, steam, and air conditioner supplier) has the smallest average value of the expected exchange rate exposure (-20.571), engage in buy transactions of currency derivatives only, and trade larger amount of swap contracts than foreign currency financing.

In summary, Panels A and B show important findings that firms with positive expected exchange rate exposure engage in larger sell transactions of currency derivatives, use larger foreign currency financing, and carry net foreign currency assets, while firms with negative expected exchange rate exposure have larger transaction amount of buying currency derivatives and carry net foreign currency debt. On the other hand, there is little difference in the swap transaction amount between firms with positive exchange rate exposure and firms with negative expected exchange rate exposure.

Viewing from the notion that firms in Table 2 have insignificant observed exchange rate exposure, the results indicate that firms use currency derivatives properly in managing exchange rate exposure. Because firms with positive expected exchange rate exposure possess foreign currency financing, they

consider management of exchange rate exposure when using foreign currency financing. In contrast, firms with negative exchange rate exposure also carry foreign currency financing, suggesting that the purpose of using foreign currency financing is not limited to the management of exchange rate exposure, and their foreign currency assets and debt are not set up to help firms manage their exchange rate exposure.

Table 3 reports results for firms with significant observed exchange rate exposure. As can be seen in Panel A, firms with positive expected exchange rate exposure carry negative observed exchange rate exposure, which suggests over-management of exchange rate exposure. While the ratios of buying and selling currency derivatives for these firms are similar to those for firms with insignificant observed exchange rate exposure (Panel A of Table 2), their magnitudes of the net position of structured forward contracts are much larger. These results suggest that firms' having negative exchange rate exposure is related to the transactions of structured forward contracts including the currency KIKO contracts.

When the expected exchange rate exposure is negative, firms on average have higher foreign currency financing ratio and swap transaction ratio than firms with insignificant observed exchange rate exposure (Panel B of Table 2). The net foreign currency debt is about twice that of firms in Panel B of Table 2. The results show that compared to firms with insignificant observed exchange rate exposure in Table 2, firms with significant observed exchange rate exposure in Table 3 do not properly use exposure-related activities to reduce the negative expected exchange rate exposure.

Because the expected exchange rate exposure estimates the effects of exchange rate changes on firm value by assuming the overall exchange rate changes as permanent, the expected exchange rate exposure tends to estimate the degree of exchange rate exposure larger than the observed exchange rate exposure does. Hence, it may be unreasonable to directly compare the magnitudes of coefficients of the expected and observed exchange rate exposures. However, the direction of the expected exchange rate exposure provides useful information on the firm's management of exchange rate exposure. The results in Tables 2 and 3 suggest that firms consider the direction of the expected exchange rate exposure in managing exchange rate exposure. In other words, the expected exchange rate exposure we estimate and

report in our paper is indeed a useful tool to separate and explain the characteristics and behavior of firms' management of exchange rate exposure.

Comparing firms with significant observed exchange rate exposure (as reported in Table 3) to firms with insignificant observed exchange rate exposure (as reported in Table 2), the following observations are obtained.

First, for firms with positive expected exchange rate exposure, firms with significant observed exchange rate exposure have greater coefficients of expected exchange rate exposure and smaller coefficients of observed exchange rate exposure than firms with insignificant observed exchange rate exposure, suggesting the possibility of over-managing exchange rate risk. While there is little difference in currency derivatives transactions, firms with significant observed exchange rate exposure tend to have more foreign currency financing, use less swap transactions, and hold a net position of structured forwards about 2.75 times that of firms with insignificant expected exchange rate exposure. These differences explain at least in part the large negative observed exchange rate exposure.

Second, when firms have negative expected exchange rate exposure, firms with significant observed exchange rate exposure have smaller coefficients of observed exchange rate exposure than firms with insignificant observed exchange rate exposure. This result confirms the notion that there is lack of an effective management of exchange rate exposure. For firms with significant exposure, the size of their buy and sell transactions of currency derivatives is the same, indicating no effects of hedging through currency derivatives. In addition, these firms possess relatively larger amount of net foreign currency debt. These results are consistent with our findings of large negative observed exchange rate exposure.

Table 4 further shows mean values of variables by year. Across the three years of 2007-2009, the expected exchange rate exposure does not exhibit a significant change, regardless of its positive or negative sign, though there are some variations in the number of firms with a positive or negative sign. In contrast, the observed exchange rate exposure reveals much greater swings across the three years. For the case of positive observed exchange rate exposure, both the number of firms and the average exchange rate exposure

decline substantially in 2008 and then increase in 2009 but to a much less extent. On the other hand, while the number of firms with negative observed exposure increases to the highest level of 299 in 2008, the average exposure declines gradually over the three years. Hence, during 2008 when the KRW/USD exchange rate soared substantially, more Korean firms experience losses in their firm values, though the average exchange rate exposure for these firms is lower than in 2007. Hence, these findings show evidence contradictory to the general belief that when exchange rates go up, exports will increase, leading to higher firm values. Table 4 also shows that more firms engage in buy- and sell-transactions of currency derivatives (*FWD-Buy* and *FWD-Sell*), structured forward contracts (*NSFWD*), and swap contracts (*SWAP*) in 2008 than in 2007 or 2009.

The overall results in Table 3 indicate that unlike the expected exchange rate exposure and variables related to hedging, the observed exchange rate exposure exhibits significant swings with respect to both the number of firms and average exchange rate exposure year by year. Accordingly, as the observed exchange rate exposure shows different changes than the expected exposure, it is necessary to examine these two exchange rate exposures separately. This notion also suggests that firms manage their exchange rate exposure by combining several widely-known tools.

3.2. *Characteristics of firms with insignificant observed exchange rate exposure*

Table 5 reports mean and median values of the expected and observed exchange rate exposures, firm characteristics, and variables representing hedging activities for firms with insignificant observed exchange rate exposure. The results are shown for three samples, a full sample and two subsamples consisting of firms with positive or negative exchange rate exposure.

Looking first at the mean values for the full sample, the expected and observed exchange rate exposure carry positive and negative signs, respectively. While the expected exchange rate exposure is a large number of 2.540, the observed exchange rate exposure has a very small absolute value of 0.020. Considering that the observed exchange rate exposure reflects the outcome of hedging, however, the finding

that the observed exchange rate exposure has a different sign than the expected exchange rate exposure suggests an over-management of exchange rate exposure.

Most variables representing firm characteristics such as export ratio (*EXPORT*), import ratio (*IMPORT*), exchange rate pass-through ratio (*PASS*), firm size (*FSIZE*), internal transactions with foreign subsidiaries (*INTTR*), and diversification index (*DIVER*) have higher ratios for firms with positive exposure than for firms with negative exposure. For example, while the mean export ratio is 26.5% for the full sample, there is substantial difference in this ratio between firms with positive expected exchange rate exposure (50.3%) and firms with negative exposure (3.5%). There is little difference in the import ratio between the two samples.

Regarding currency derivatives, firms with positive (negative) exchange rate exposure on average engage in more transactions of selling (buying) currency derivatives. For the full sample, firms use more transactions of selling currency derivatives than buying such contracts. The net position of structured forwards is shown to be a net sell position for the three samples; for example, firms with positive expected exchange rate exposure hold about 0.4% of such contracts relative to firm size, while firms with negative expected exposure carry about 0.1% of such contracts. Foreign currency financing amount is about 3.6% of firm size for firms with positive expected exchange rate exposure and about 2.3% for firms with negative expected exchange rate exposure. There is little difference in the magnitude of swap contracts based on the direction of expected exchange rate exposure. While firms in the full sample own about 1% of net foreign currency debt relative to firm size, firms with positive expected exchange rate exposure have net foreign currency assets but firms with negative expected exchange rate exposure possess net foreign currency debt.

Table 6 reports mean values of expected (EX) and observed (OB) exchange rate exposures for several categories classified by different relations between these two exposures by year. Similarly to Table 5, the results in Table 6 are for firms with insignificant observed exchange rate exposure which thus manage their exposure relatively well. The categories of +EX>+OB and -EX<-OB are where observed

exposure is less than expected exposure (in absolute value), hence indicating an effective hedging. The total number of firms belonging to these two categories is 136 (36.3% out of 375) in 2007, 107 (29.4% out of 364) in 2008, and 155 (40.3% out of 385) in 2009. It is particularly interesting to see that the number of firms for the category of +EX>+OB declines substantially to 32 in 2008, less than half of 69 in 2007. This observation of a big swing in the number of firms is attributed to the fact that in 2008 a large number of firms carry significant observed exchange rate exposures due to the large increases in exchange rate and hence these firms are excluded from the list in Table 6.

The categories of +EX, -OB and -EX, +OB are where firms over-manage their exchange rate exposure. While the number of firms for +EX, -OB gradually increases, the number of firms for -EX, +OB declines substantially to 26 in 2008 (from 101 in 2007). Hence, as the exchange rate increases significantly in 2008, the degree of over-management of exchange rate exposure by Korean firms with negative expected exposure seems to have been eased to some extent.

The findings in Table 6 indicate that the characteristics of the observed exchange rate exposure reflected in the firms' stock returns can vary significantly depending on the direction of the expected exchange rate exposure. Similarly to those reported in earlier tables, the results in Table 6 also suggest that it will be necessary to investigate not only the observed and expected exposures separately but also the directions of these exposures as well.

3.3 *Pearson correlation coefficients*

Table 7 shows Pearson correlation coefficients of variables of interest for firms with insignificant observed exchange rate exposure. Panels A and B report results for firms whose expected exchange rate exposure is positive and negative, respectively.

The results in Panel A show that the expected exchange rate exposure is highly positively correlated with *EP* (export ratio), *IT* (internal transaction ratio), and *FS* (sell transactions of currency derivatives). The highly positive coefficient (0.645) of *EP* indicates that a firm's export ratio explains a large portion of

the positive expected exchange rate exposure, whereas the highly positive coefficient (0.229) of *FS* suggests that the sell transactions of currency derivatives are related to the firms' hedging activities. *IP* (import ratio) is highly positively correlated with *PT* (pass-through ratio) with the correlation coefficient of 0.837, implying that firms using relatively large amount of imported goods for their production tend to pass through the effects of exchange rate exposure to their pricing policies. The highly positive coefficient of 0.243 between *FS* and *FL* (buy transactions of currency derivatives) implies that currency derivatives are not merely used for the purpose of managing firms' exchange rate exposure. The highly positive coefficient of 0.503 between *FF* (foreign currency financing) and *FD* (net foreign currency debt) suggests that foreign currency financing is done primarily with foreign currency debt.¹²

Turning to results in Panel B, the expected exchange rate exposure is highly negatively corrected with *IP* and *PT*. Considering the negative sign of expected exposure, greater expected exchange rate exposure is associated with higher import ratios and more exchange rate pass-through. Unlike firms with positive expected exchange rate exposure reported in Panel A, *EP* is highly positively correlated with *SF* (net sell position of structured forwards) (corr. coeff. = 0.347). Hence, firms whose firm value declines in response to an increase in the exchange rate tend to take greater short positions in (that is, selling) structured forwards. This evidence suggests that firms use structured forwards solely based on the firms' exporting activities instead of considering the firms' overall positions in foreign currency.¹³ *FD* and *SW* (swap transactions) are highly positively correlated to each other (correlation coefficient = 0.340), showing that firms with relatively large amount of net foreign currency debt also engage in large transactions of swap contracts.

In summary, comparing the results in Panels A and B of Table 7, the expected exchange rate

¹² It is worth noting that foreign currency debt also includes certain obligations (e.g., credit and accounts payable) that is not directly related to foreign currency financing. For example, if a firm imports raw materials from a foreign supplier through usance (e.g., credit), the amount can also be treated as foreign currency debt.

¹³ During our sample period, the majority of structured currency forward contracts are the currency KIKO option contracts.

exposure and the variables necessary for the estimation of expected exchange rate exposure have in general the expected relationships postulated in equation (4), but the relationships among these variables vary significantly based on the direction (positive or negative) of the expected exchange rate exposure. Viewing from these findings, if one analyzes the variables altogether without taking into account the direction of the expected exchange rate exposure, the analysis would not produce meaningful results on the relationship between the expected exchange rate exposure and its related variables.

3.4. *Regression results of difference in exchange rate exposures (EDIF)*

Table 8 presents regression results of difference in exchange rate exposure (*EDIF*) as dependent variable estimated from equation (7) by two subgroups of firms based on the direction of the expected exchange rate exposure and a pooled group. It is important to note that because *EDIF* is computed as the difference between the expected and observed exchange rate exposure, *EDIF* will have a larger value when the expected exchange rate exposure is positive and the observed exchange rate exposure is smaller (that is, the more effective the hedging activity is). In contrast, when the expected exchange rate exposure is negative and the observed exchange rate exposure is larger, *EDIF* will have a smaller value (or a larger absolute value). In other words, when the expected exchange rate exposure is positive, the larger the *EDIF* is, the more effective the hedging activity is. When the expected exchange rate exposure is negative, the smaller the *EDIF* is (that is, the larger the absolute value of the negative *EDIF*), the more effective the hedging activity is.

If the signs of the observed exchange rate exposure are different from those of the expected exchange rate exposure, it would be difficult to explain the relationship between *EDIF* and the hedging activities in a consistent manner, but one still should be able to examine the relationship between firms' exchange rate exposure and activities that firms use to manage their exposure. In addition, as we postulate in this paper, when the observed exchange rate exposure is not significant, the analysis of the relationship between *EDIF* and its related variables allows us to examine whether firms properly use various hedging

activities. Accordingly, the results in Table 8 can be interpreted as the evidence for firms that have performed their hedging activities properly.

It is worth noting that the regression model (7) does not include variables for firms' swap transactions (SWAP) or net foreign currency debt (NFCDEBT). Because both swap transaction amount and foreign currency debt are used to measure a firm's net amount of foreign currency financing (NFCFIN), we exclude them to avoid a double counting problem of these variables.

For firms with positive expected exchange rate exposure, *EDIF* is significantly positively related to *FWD-Sell*, *NFCFIN*, and *INTTR* at least at the 5% level, suggesting effective managements of these activities in reducing exchange rate exposure. On the other hand, *NSFWD* is positively related to *EDIF*, but the relationship is not significant at the 10% level. This finding offers interesting evidence that like exporting firms, firms whose values decline in response to a decrease in the exchange rate do not gain much from the usage of the structured currency derivatives contracts in managing exchange rate exposure.

Viewing from the notion that the positive expected exchange rate exposure means taking a long position in foreign currency contracts, this result is difficult to interpret because the short positions in structured forwards should result in the reduction in firms' exchange rate exposure. One possible explanation is that among the firms that use structured forwards, some firms use such contracts without considering their overall positions in the exchange rate exposure. Another possibility is that because the structured forward contracts (including KIKO contracts) Korean firms use are designed to be cancelled when the exchange rate declines below a certain level, these types of derivatives are not effective in managing exchange rate exposure resulting from long positions of a foreign currency. If the market responds by expecting this notion, then the usage of the structured forwards would have little effect on the exchange rate exposure.

The results for firms with negative expected exchange rate exposure show that *NFCFIN* has a positive and significant (at the 10% level) effect on *EDIF*, hence reflecting the positive relationship between exchange rate exposure difference and net foreign currency debt. *NSFWD* is significantly positively

related to *EDIF* at the 1% level, hence increasing the exchange rate exposure. This result is not surprising because the net position of structured forwards has a similar benefit-loss structure to the sell transaction of currency derivatives.

EDIF is significantly negatively related to *PASS* at the 1% level; hence, a firm's pass-through activity works in the direction of reducing the difference in exchange rate exposure, or decreasing the observed exchange rate exposure. This evidence on the effectiveness of the firm's pass-through activity in reducing exchange rate exposure is different from that for firms with positive expected exchange rate exposure. This finding indicates that similarly to importing firms, firms whose values decline when the exchange rate increases manage their exchange rate exposure by passing through exchange rate changes to product pricing.

FWD-Buy carries a positive but insignificant coefficient at any level. Hence, the buy transactions of currency derivatives have little effect on the reduction of exchange rate exposure. Considering the benefit-loss structure of the buy transactions of currency derivatives in response to exchange rate changes, this result is somewhat surprising. This evidence can be explained by the notions that the size of the buy transactions of currency derivatives is not large enough to affect firm values and/or that some firms engage in buy transactions of currency derivatives for purposes other than managing exchange rate exposure.

For a comparison purpose, Table 8 also reports the regression estimates for the pooled group by ignoring the direction of the expected exchange rate exposure. While some variables show regression estimates for the pooled group similar to those for the two subgroups, a few other variables such as *NSFWD*, *NFCFIN*, *PASS*, *INTTR*, and *DIVER* reveal noticeable differences in their regression estimates between the pooled group and the two subgroups. For example, *NFCFIN* has a positive and insignificant regression coefficient for the pooled group but positive and significant coefficients for the two subgroups. Hence, if relying on the regression results for the pooled group, one would be misled to the conclusion of no significant effect of a firm's foreign debt financing. Similarly, the insignificant regression coefficient of *PASS* for the pooled group would undermine the evidence that firms with negative expected exposure

manage their exposure by the exchange-rate pass-through activity, as shown for the subgroup of firms with negative expected exposure. The positive and significant regression coefficient of *FSIZE* for the pooled sample shows that the larger the firm size, the more active the firm's management of exchange rate exposure. This finding offers evidence supporting the notion that larger firms tend to be more active hedgers against their exchange rate exposures, whose evidence is consistent with the findings in the existing literature.

The regression results in Table 8 offer confirmatory evidence that the effectiveness of several hedging activities firms use for their exposure management varies depending upon the direction of the firms' expected exchange rate exposure. Relying simply on the regression results from a pooled sample without considering this notion would likely yield very limited and unreliable empirical implications. Our results strongly suggest that it is necessary to consider the characteristics (directions) of firms' exchange rate exposure in order to properly analyze the relationship between exchange rate exposure and hedging activities.

3.5. *Robustness tests*

We conduct two additional tests in order to ensure the robustness of our main results. First, our analyses so far have been performed for the sample that excludes firms whose profit margins are below 25% of the average profit margin of all firms. This considers the notion that because the model for the expected exchange rate exposure is developed under the assumption of firms' profit maximization, it would be difficult to give a meaningful interpretation to the expected exchange rate exposure of firms with low operating performance. Furthermore, this also considers the technical issue that when a firm's profit margin (r) in equation (5) is extremely low, the value of the expected exchange rate exposure will be computed as a significantly amplified value.

It is ideal to define ex-ante the minimum level of a firm's profit margin in order to use properly the computed expected exchange rate exposure in the analysis of exchange rate exposure. Because the model for the expected exchange rate exposure assumes that the firm's profit margin is given, however, it is

impossible to define the appropriate level of profit margin in the model. In order to help understand the level of profit margin that can give a meaningful interpretation to the computed expected exchange rate exposure, we examine the relationship between exchange rate exposure and several hedging activities for exposure management at various levels of profit margin. For this purpose, we first construct two subsamples from all sample firms by excluding firms whose profit margins are less than 10% and 15% of the average of all firms. We then estimate regression model (7) for the two subsamples. The average profit margin for the sample firms is 7.38% for the entire three-year sample period and 7.06%, 7.86%, and 7.37% in 2007, 2008, and 2009, respectively.

Table 9 shows the results. Overall, the results from sample firms with different thresholds of profit margin offer confirmatory evidence to that reported earlier. Firms with positive expected exchange rate exposure reveal a reduction in exchange rate exposure resulting from the sell transactions of currency derivatives (*FWD-Sell*). In contrast, firms with negative expected exchange rate exposure do not show such a reduction in exchange rate exposure with the buy transactions of currency derivatives, as evidenced by the insignificant regression coefficient of *FWD-Buy*. For firms with negative expected exchange rate exposure, *NSFWD* is significantly positively related to exchange rate exposure difference, indicating that the short position of structured forwards brings in a negative effect in the reduction of exchange rate exposure. On the other hand, these firms manage their exchange rate exposure using transfer pricing activities. *INTTR* is significantly positively related to the reduction of exchange rate exposure for firms with both positive and negative expected exchange rate exposures. The regression coefficient of *DIVER* is negative but significant (at the 10% level) only for firms with positive expected exposure. The results from the second sample (by excluding firms whose profit margins are less than 15% of the average of all firms) are similar to those from the first sample in Table 9.¹⁴

Second, our analyses employed daily stock returns for sample firms and the value-weighted KOSPI

¹⁴ Though not reported in our paper, the analyses using sample firms with different threshold levels of profit margin of higher than 15% show results similar to those reported in this paper.

market index in estimating observed exchange rate exposures in regression equation (6). Bodnar and Wong (2003) show that both a choice of market portfolios and the return measurement horizon make impacts on estimates of exchange rate exposure for U.S. firms.¹⁵ It is conceivable that the usage of a value-weighted market index would reflect relatively more weights for large firms, the usage of an equal-weighted market portfolio would treat large firms and small firms equally. In order to examine whether our main results are sensitive to different market index and return horizon, we estimate regression equation (6) using returns of the equal-weighted KOSPI market index and monthly stock returns for sample firms and the market index in replace of daily stock returns. Table 10 present the regression estimates with alternative market index and return horizon. While the signs and significance levels of two control variables, *DIVER* and *FSIZE*, vary to some degree relative to those in Table 8, the regression estimates of main testing variables for firms' hedging activities remain qualitatively identical to those in Table 8. Firms with positive expected exchange rate exposure exhibit a decrease in exchange rate exposure through selling currency derivatives (*FWD-Sell*), the issuance of foreign currency debt (*NFCFIN*), and internal transactions with foreign subsidiaries (*INTTR*) in the regressions with the equal-weighted market index or monthly stock returns. On the contrary, firms with negative expected exchange rate exposure reduce their exchange rate exposures through exchange rate pass-through activities (*PASS*). Interestingly, similar to the regression estimate in Table 8, *NSFWD* carries a significant and positive regression coefficient for firms with negative expected exchange rate exposure. This result indicates that short positions of structured forwards increase the exchange rate exposure further rather than helping reduce the exposure. The overall results in Table 10 confirm that the main results of our study remain robust to alternative market index and return horizon.

¹⁵ Bodnar and Wong (2003) note that while value-weighted market returns provide exposure estimates with wrong signs, equally-weighted market portfolios correct this problem.

4. Summary and conclusions

In examining the relationship between firms' exchange rate exposures and their various hedging activities for exposure management, we take a different approach from those in the existing literature. We measure separately the expected exchange rate exposure that is determined by firms' basic business activities and the observed exchange rate exposure that reflects the effectiveness of firms' activities in managing their exposures.

Employing all Korean non-financial firms, we show that firms with positive expected exchange rate exposures obtain the benefits of reducing their exchange rate exposures through the sell transactions of currency derivatives, internal transactions with foreign subsidiaries, and foreign currency financing. In contrast, firms with negative exchange rate exposures are able to reduce their exchange rate exposures only through passing through exchange rate changes to their product pricing. The analysis of firms with significant observed exchange rate exposures offers evidence supporting the widely-believed notion in the Korean markets that structured currency forwards contracts (including the currency KIKO contracts) are the key factor that worsens firms' exchange rate exposures. We also find that Korean firms in general properly use several operating and financial activities in managing their exchange rate exposures. A good number of firms are, however, shown to fail to use such activities properly, and some firms over-hedge their exchange rate exposures, indicating that it is necessary to take a systematic approach for the management of exchange rate exposure.

The results of our paper strongly suggest that in order to assess the effectiveness of operating and financial hedging activities for exposure management, it is imperative to consider the underlying characteristics (e.g., direction) of firms' exchange rate exposures. Furthermore, the effectiveness of hedging activities in the financial markets can vary depending on the conditions in the product market (e.g., export and import ratios and operating profitability) that firms face before engaging such hedging activities. Therefore, relying simply on the regression results of observed stock returns without considering these notions would likely yield very limited and unreliable empirical evidence on the effectiveness of hedging

activities.

The new approach to measuring expected exchange rate exposure and other research methodologies employed in this paper would offer rich applications to the research of similar issues for many other countries where local firms are exposed to high exchange rate risk but whose empirical evidence is currently very scarce. In this endeavor, future research could also examine other currencies and/or a broader index of currencies.

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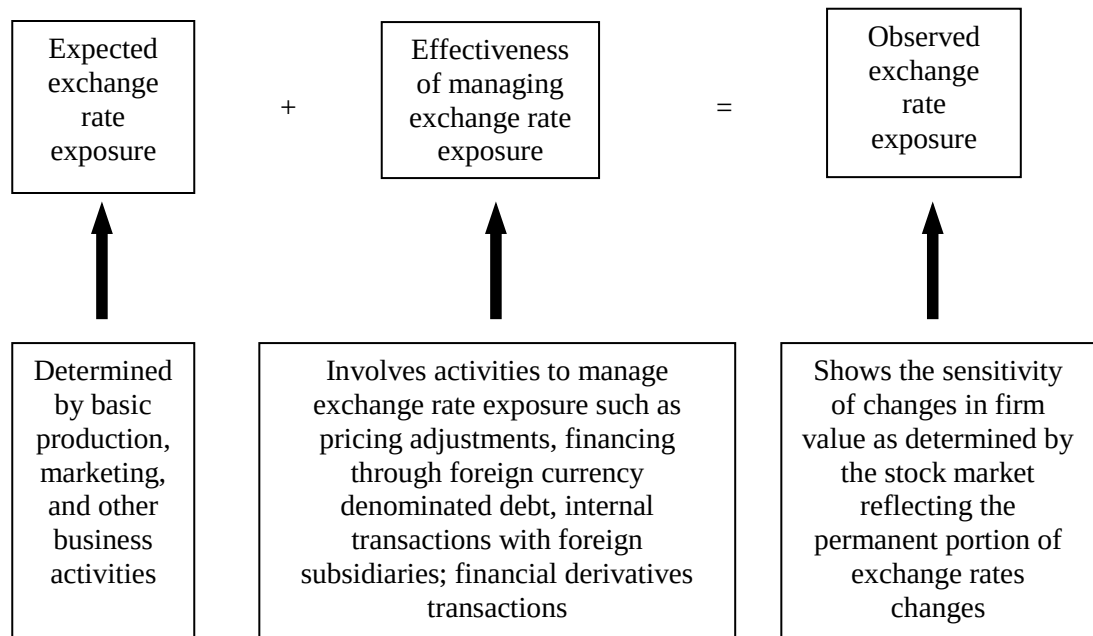


Figure 1. Two types of exchange rate exposure

Table 1

The relationships between expected exchange rate exposure and effectiveness of hedging activities given smaller and statistically insignificant observed exchange rate exposure

If observed ER exposure is	and expected ER exposure is	Then EDIF will be	Then a hedging activity will be regarded as
Smaller/insignificant	Positive	A larger value	More effective
Smaller/insignificant	Negative	A smaller value (or a larger absolute value)	More effective

Table 2

Exchange rate exposure and related variables for firms with insignificant observed exposure

KSIC	Expected ER exposure	Observed ER exposure	FWD- Buy	FWD- Sell	NS- FWD	FCFIN	SWAP	NFC- DEBT	No. of firms
<i>Panel A. Firms with positive expected exchange rate exposure</i>									
10	2.935	-0.009	0.002	0.005	0.000	0.082	0.009	0.044	16
13	12.794	-0.047	0.001	0.119	0.000	0.121	0.000	-0.074	10
17	3.720	-0.279	0.003	0.000	0.000	0.078	0.010	0.116	8
20	6.529	0.008	0.002	0.012	0.005	0.041	0.013	0.021	62
21	2.247	0.032	0.009	0.000	0.000	0.012	0.010	-0.026	10
22	8.762	0.067	0.006	0.021	0.005	0.016	0.016	-0.052	25
23	3.225	0.034	0.002	0.030	0.000	0.034	0.003	0.008	7
24	2.912	-0.066	0.000	0.030	0.007	0.053	0.064	0.043	35
26	10.468	0.049	0.000	0.005	0.000	0.028	0.002	-0.053	26
28	9.764	-0.117	0.002	0.023	0.001	0.020	0.013	-0.025	16
29	6.433	-0.063	0.002	0.078	0.006	0.014	0.007	-0.050	30
30	8.706	0.000	0.000	0.017	0.009	0.022	0.004	-0.037	46
31	11.904	-0.258	0.101	0.884	0.000	0.008	0.001	-0.102	14
41	5.203	-0.041	0.019	0.066	0.000	0.006	0.009	-0.026	30
46	13.449	0.013	0.012	0.012	0.000	0.035	0.001	0.004	21
Other	7.161	-0.055	0.030	0.033	0.003	0.052	0.014	0.019	64
Total	7.288	-0.031	0.011	0.057	0.004	0.036	0.013	-0.009	420
<i>Panel B. Firms with negative expected exchange rate exposure</i>									
10	-1.136	-0.064	0.001	0.000	0.000	0.086	0.025	0.095	40
14	-1.482	-0.107	0.000	0.000	0.000	0.006	0.001	0.011	13
17	-1.814	-0.148	0.000	0.000	0.008	0.049	0.055	0.093	15
19	-1.975	-0.197	0.000	0.000	0.000	0.008	0.000	-0.025	8
20	-1.915	-0.014	0.000	0.002	0.000	0.012	0.006	0.014	45
21	-1.352	0.044	0.002	0.000	0.000	0.003	0.008	0.017	56
23	-1.403	-0.075	0.000	0.000	0.000	0.014	0.004	0.017	17
24	-1.701	-0.007	0.493	0.005	0.019	0.076	0.006	0.059	19
25	-0.480	0.203	0.014	0.002	0.000	0.053	0.000	0.064	11
26	-6.531	0.142	0.059	0.012	0.000	0.019	0.033	0.009	7
35	-20.571	-0.026	0.013	0.000	0.000	0.004	0.010	0.023	13
41	-0.737	-0.044	0.000	0.000	0.000	0.002	0.002	-0.001	41
46	-0.649	0.089	0.000	0.000	0.000	0.010	0.018	-0.010	14
47	-0.431	0.119	0.000	0.000	0.000	0.004	0.013	0.013	27
49	-2.720	-0.056	0.000	0.000	0.000	0.017	0.018	0.016	9
58	-0.806	0.123	0.000	0.000	0.000	0.000	0.001	0.000	16
Other	-1.914	-0.045	0.001	0.002	0.000	0.025	0.009	0.027	83
Total	-2.054	-0.009	0.024	0.001	0.001	0.023	0.011	0.027	434

This table reports mean values of expected and observed exchange rate exposures and variables related to hedging activities by industry for firms with statistically insignificant observed exchange rate exposure. *FWD-Buy* and *FWD-Sell* are buy- and sell-transaction amount of currency derivatives, respectively. *NSFWD* is the net position in structured forward contracts. *FCFIN* is foreign currency financing amount. *SWAP* represents contracts of currency swaps and currency interest swaps. *NFCDEBT* is net foreign currency debt.

Table 3

Exchange rate exposure and related variables for firms with significant observed exposure

KSIC	Expected ER exposure	Observed ER Exposure	FWD- Buy	FWD- Sell	NS- FWD	FCFIN	SWAP	NFC- DEBT	No. of firms
<i>Panel A. Firms with positive expected exchange rate exposure</i>									
20	4.557	-0.206	0.003	0.015	0.008	0.066	0.014	0.042	15
24	3.387	-0.385	0.004	0.018	0.018	0.080	0.014	0.084	16
26	14.119	0.080	0.000	0.002	0.000	0.010	0.003	-0.047	14
28	7.475	-0.224	0.008	0.048	0.000	0.041	0.029	-0.008	14
29	4.870	-0.305	0.008	0.328	0.023	0.005	0.000	-0.065	9
30	16.527	-0.205	0.000	0.053	0.017	0.021	0.008	-0.048	10
41	3.189	0.006	0.003	0.004	0.000	0.011	0.004	-0.008	8
46	13.092	-0.738	0.046	0.023	0.000	0.046	0.007	0.021	11
Other	8.173	-0.070	0.016	0.063	0.015	0.067	0.007	0.005	53
Total	8.244	-0.186	0.011	0.056	0.011	0.049	0.010	0.004	150
<i>Panel B. Firms with negative expected exchange rate exposure</i>									
10	-0.810	-0.290	0.009	0.001	0.000	0.120	0.041	0.145	13
17	-1.572	-0.225	0.000	0.000	0.000	0.068	0.056	0.112	7
20	-1.993	-0.339	0.000	0.001	0.000	0.039	0.039	0.078	11
21	-0.753	-0.213	0.000	0.000	0.000	0.001	0.018	0.034	7
24	-2.563	-0.164	0.002	0.012	0.000	0.079	0.010	0.065	9
41	-0.753	-0.322	0.000	0.000	0.000	0.000	0.003	0.002	17
46	-0.634	-0.336	0.005	0.000	0.000	0.027	0.028	0.030	13
Other	-5.075	-0.081	0.000	0.003	0.000	0.020	0.012	0.028	43
Total	-2.592	-0.211	0.002	0.002	0.000	0.037	0.021	0.050	120

This table reports mean values of expected and observed exchange rate exposures and variables related to hedging activities by industry for firms with statistically significant (at least at the 10% level) observed exchange rate exposure. *FWD-Buy* and *FWD-Sell* are buy- and sell-transaction amount of currency derivatives, respectively. *NSFWD* is the net position in structured forward contracts. *FCFIN* is foreign currency financing amount. *SWAP* represents contracts of currency swaps and currency interest swaps. *NFCDEBT* is net foreign currency debt.

Table 4

Characteristics of expected and observed exchange rate exposures and related variables by year

Variable		Year 2007		Year 2008		Year 2009	
		No. of firms	Mean	No. of firms	Mean	No. of firms	Mean
Expected ER Exposure	Positive	178	7.935	196	7.238	196	7.482
	Negative	197	-2.382	168	-2.162	189	-1.957
Observed ER Exposure	Positive	201	0.509	65	0.100	155	0.185
	Negative	174	-0.577	299	-0.223	230	-0.177
FWD	Buy	29	0.115	47	0.251	29	0.060
	Sell	52	0.161	69	0.198	63	0.176
NSFWD		14	0.095	18	0.091	8	0.082
FCFIN		145	0.060	168	0.079	181	0.082
SWAP		16	0.044	27	0.063	14	0.028
NFCDEBT		174	0.068	172	0.093	173	0.095
No. of firms		375		364		385	

This table reports mean values of expected and observed exchange rate exposures and variables related to hedging activities by year. *FWD-Buy* and *FWD-Sell* are buy- and sell-transaction amount of currency derivatives, respectively. *NSFWD* is the net position in structured forward contracts. *FCFIN* is foreign currency financing amount. *SWAP* represents contracts of currency swaps and currency interest swaps. *NFCDEBT* is net foreign currency debt.

Table 5

Summary statistics of variables for firms with insignificant observed exchange rate exposure

Variables	Full sample		Firms with positive ER exposure		Firms with negative ER exposure	
	Mean	Median	Mean	Median	Mean	Median
<i>Expected ER exposure</i>	2.540	-0.032	7.288	5.420	-2.054	-0.964
<i>Observed ER exposure</i>	-0.020	-0.048	-0.031	-0.049	-0.009	-0.048
<i>EXPORT</i>	0.265	0.112	0.503	0.497	0.035	0.005
<i>IMPORT</i>	0.156	0.119	0.159	0.125	0.153	0.110
<i>PASS</i>	0.323	0.307	0.350	0.329	0.297	0.269
<i>INTTR</i>	0.075	0.001	0.127	0.036	0.024	0.000
<i>DIVER</i>	0.261	0.059	0.271	0.065	0.251	0.057
<i>FSIZE</i>	19.724	19.321	19.976	19.463	19.480	19.191
<i>FWD-Buy</i>	0.018	0.000	0.011	0.000	0.024	0.000
<i>FWD-Sell</i>	0.029	0.000	0.057	0.000	0.001	0.000
<i>NSFWD</i>	0.002	0.000	0.004	0.000	0.001	0.000
<i>FCFIN</i>	0.029	0.000	0.036	0.003	0.023	0.000
<i>SWAP</i>	0.012	0.000	0.013	0.000	0.011	0.000
<i>NFCDEBT</i>	0.010	0.000	-0.009	-0.009	0.027	0.000

This table reports mean and median values of expected and observed exchange rate exposures, firm characteristics, and variables related to hedging activities for firms with statistically insignificant observed exchange rate exposure. *IMPORT* (import ratio) and *PASS* (exchange rate pass-through ratio) are estimated by relating firm's sales composition to imported input shares of sector sales and item-by-item transfer pricing, respectively. *INTTR* is the internal transactions with foreign subsidiaries. *DIVER* is diversification index. *FSIZE* is firm size. *FWD-Buy* and *FWD-Sell* are buy- and sell-transaction amount of currency derivatives, respectively. *NSFWD* is the net position in structured forward contracts. *FCFIN* is foreign currency financing amount. *SWAP* represents contracts of currency swaps and currency interest swaps. *NFCDEBT* is net foreign currency debt.

Table 6

Expected and observed exchange rate exposures based on their relations by year

Category	Year 2007			Year 2008			Year 2009		
	No. of firms = 375			No. of firms = 364			No. of firms = 385		
	No. of firms	EX	OB	No. of firms	EX	OB	No. of firms	EX	OB
+EX>+OB	69	8.126	0.368	32	6.096	0.068	67	6.995	0.143
-EX<-OB	67	-2.612	-0.363	75	-2.501	-0.117	88	-2.293	-0.131
+EX<+OB	2	0.227	0.870	0	n/a	n/a	4	0.046	0.218
-EX>-OB	10	-0.287	-0.726	5	-0.100	-0.163	6	-0.080	-0.196
+EX, -OB	75	7.975	-0.421	83	6.800	-0.112	88	7.651	-0.136
-EX,+OB	101	-1.941	0.398	26	-1.210	0.071	56	-1.708	0.140

This table reports mean values of expected (EX) and observed (OB) exchange rate exposures for several categories classified by different relations between these two exposures by year for firms with statistically insignificant observed exchange rate exposure. Expected exchange rate exposure reflects a firm's exposure level related to its basic business activities prior to its usage of any hedging activity, and observed exchange rate exposure reflects the effects of the firm's other hedging activities for exposure management as well.

Table 7

Pearson correlation coefficients of variables for firms with insignificant observed exchange rate exposure

<i>Panel A. Firms with positive expected exchange rate exposure</i>														
Var	EX	OX	EP	IP	PT	IT	DV	FV	FL	FS	SF	FF	SW	FD
OX	-0.005	1.000												
EP	0.645	-0.040	1.000											
IP	-0.057	-0.087	0.406	1.000										
PT	-0.097	-0.054	0.277	0.837	1.000									
IT	0.349	0.149	0.354	0.035	-0.059	1.000								
DV	-0.079	0.002	-0.092	-0.085	-0.133	-0.079	1.000							
FV	-0.066	-0.052	0.150	0.194	0.048	-0.011	0.260	1.000						
FL	0.129	-0.045	0.064	-0.022	-0.066	-0.015	0.051	0.068	1.000					
FS	0.229	-0.158	0.334	0.057	-0.001	-0.062	-0.062	0.277	0.243	1.000				
SF	0.063	0.038	0.059	0.021	0.038	0.009	-0.032	-0.085	-0.015	-0.028	1.000			
FF	0.111	-0.092	0.097	0.144	0.148	0.020	-0.070	-0.089	-0.027	-0.047	0.019	1.000		
SW	-0.032	0.149	-0.041	0.066	0.097	-0.055	-0.013	-0.031	-0.011	-0.033	0.010	-0.025	1.000	
FD	-0.176	-0.070	-0.145	0.227	0.206	-0.118	0.069	0.240	-0.015	-0.266	-0.042	0.503	0.089	1.000
<i>Panel B. Firms with negative expected exchange rate exposure</i>														
Var	EX	OX	EP	IP	PT	IT	DV	FV	FL	FS	SF	FF	SW	FD
OX	-0.013	1.000												
EP	0.094	-0.107	1.000											
IP	-0.764	-0.054	0.379	1.000										
PT	-0.608	-0.085	0.407	0.874	1.000									
IT	0.033	-0.037	0.264	0.111	0.124	1.000								
DV	0.072	-0.038	-0.028	-0.131	-0.140	0.003	1.000							
FV	0.016	-0.028	-0.122	-0.061	-0.151	0.021	0.029	1.000						
FL	0.013	-0.036	-0.021	-0.004	0.023	-0.005	0.041	-0.025	1.000					
FS	0.040	0.047	0.150	0.026	0.070	0.115	-0.082	-0.045	-0.005	1.000				
SF	0.021	-0.013	0.347	0.120	0.138	-0.010	-0.043	-0.058	-0.004	-0.011	1.000			
FF	0.053	-0.057	0.068	-0.041	0.174	0.009	0.002	-0.121	0.194	-0.008	0.094	1.000		
SW	0.027	-0.084	0.024	-0.024	0.009	-0.054	-0.061	0.202	-0.016	-0.027	-0.026	0.104	1.000	
FD	0.026	-0.047	-0.050	-0.071	0.099	0.021	-0.032	0.007	0.154	-0.033	0.043	0.826	0.340	1.000

EX = expected exchange rate exposure; OX = observed exchange rate exposure; EP = export ratio; IP = import ratio; PT = pass-through ratio; IT = internal transactions with foreign subsidiaries; DV = diversification index; FV = firm size; FL = buy transactions of currency derivatives; FS = sell transactions of currency derivatives; SF = net position of structured forwards; FF = Foreign currency financing; SW = swap contracts; FD = net Foreign currency debt

Table 8

Regression results of difference in exchange rate exposure

Variables	Positive expected ER exposure	Negative expected ER exposure	Pooled
<i>Constant</i>	11.862*** (2.78)	-2.584 (-1.25)	-5.702* (1.72)
<i>FWD-Sell</i>	12.223*** (4.76)		14.781*** (5.74)
<i>FWD-Buy</i>		0.044 (0.60)	0.128 (0.26)
<i>NSFWD</i>	17.572 (1.33)	9.914*** (2.75)	18.549* (1.70)
<i>NFCFIN</i>	4.207** (2.02)	1.915* (1.83)	4.340 (1.56)
<i>PASS</i>	-0.483 (-0.12)	-7.320*** (-3.64)	-0.603 (-0.18)
<i>INTTR</i>	10.144*** (5.08)	0.606 (1.09)	13.131*** (6.50)
<i>DIVER</i>	-0.273 (-0.30)	-0.509* (-1.88)	-0.079 (-0.14)
<i>FSIZE</i>	-0.492** (-2.40)	0.200** (2.02)	0.261* (1.67)
<i>IDUMMY</i>	Yes	Yes	Yes
<i>YDUMMY</i>	Yes	Yes	Yes
Adjusted R ²	0.281	0.771	0.436
No. of firms	420	434	854

The dependent variable is *EDIF*, the difference between expected and observed exchange rate exposures. *FWD-Buy* and *FWD-Sell* are buy- and sell-transaction amount of currency derivatives, respectively. *NSFWD* is the net position in structured forward contracts. *NFCFIN* is net foreign currency financing amount. *PASS* is exchange rate pass-through ratio. *INTTR* is the internal transactions with foreign subsidiaries. *DIVER* is diversification index. *FSIZE* is firm size. *IDUMMY* and *YDUMMY* are dummy variables for industry and year, respectively, whose results are not reported here for brevity's sake. t-statistics are in parentheses.

* denotes significance at the 10% level.

** denotes significance at the 5% level.

*** denotes significance at the 1% level.

Table 9

Robustness test using different thresholds of profit margin

Variables	Sample by excluding firms whose profit margins are less than 10% of the average for all firms		Sample by excluding firms whose profit margins are less than 15% of the average for all firms	
	Firms with positive expected ER exposure	Firms with negative expected ER exposure	Firms with positive expected ER exposure	Firms with negative expected ER exposure
<i>Constant</i>	12.321*** (2.59)	-4.755* (-1.70)	7.915 (1.17)	-3.558 (-1.51)
<i>FWD-Sell</i>	10.218*** (3.74)		10.700*** (3.73)	
<i>FWD-Buy</i>		0.115 (0.88)		0.116 (1.12)
<i>NSFWD</i>	9.696 (0.72)	15.526*** (3.01)	13.711 (1.02)	16.585*** (3.22)
<i>NFCFIN</i>	7.062** (2.01)	1.775* (1.71)	4.028* (1.81)	1.580 (1.00)
<i>PASS</i>	-0.440 (-0.05)	-10.091*** (-3.22)	-2.912 (-0.44)	-10.122*** (-4.32)
<i>INTTR</i>	7.001*** (2.52)	1.280* (1.72)	6.643*** (2.69)	1.039 (1.48)
<i>DIVER</i>	-2.820* (-1.74)	-0.408 (-1.25)	-2.821* (-1.82)	-0.331 (-1.10)
<i>FSIZE</i>	-0.343 (-0.90)	0.349*** (2.71)	-0.092 (-0.26)	0.238*** (2.68)
<i>IDUMMY</i>	Yes	Yes	Yes	Yes
<i>YDUMMY</i>	Yes	Yes	Yes	Yes
Adjusted R ²	0.166	0.545	0.188	0.704
No. of firms	461	471	448	462

The dependent variable is *EDIF*. *FWD-Buy* and *FWD-Sell* are buy- and sell-transaction amount of currency derivatives, respectively. *NSFWD* is net position in structured forward contracts. *NFCFIN* is net foreign currency financing amount. *PASS* is exchange rate pass-through ratio. *INTTR* is internal transactions with foreign subsidiaries. *DIVER* is diversification index. *FSIZE* is firm size. *IDUMMY* and *YDUMMY* are dummy variables for industry and year, respectively, whose results are not reported here for brevity's sake. t-statistics are in parentheses.

* denotes significance at the 10% level.

** denotes significance at the 5% level.

*** denotes significance at the 1% level.

Table 10

Robustness tests using equally-weighted market index and monthly stock returns

Variables	With equally-weighted market index			With monthly stock returns		
	Positive exp. ER exposure	Negative exp. ER exposure	Pooled	Positive exp. ER exposure	Negative exp. ER exposure	Pooled
<i>Constant</i>	7.857 (1.19)	-3.784 (-1.63)	-9.478** (-2.06)	7.480 (1.12)	-3.502 (-1.44)	-9.596** (-2.06)
<i>FWD-Sell</i>	10.704*** (3.80)		14.052*** (5.14)	10.666*** (3.80)		13.977*** (5.14)
<i>FWD-Buy</i>		0.104 (1.01)	0.203 (0.39)		0.126 (1.16)	0.226 (0.43)
<i>NSFWD</i>	13.173 (1.05)	16.354*** (3.17)	16.707* (1.71)	13.441 (1.08)	17.407*** (3.22)	17.042* (1.77)
<i>NFCFIN</i>	4.057* (1.84)	1.575 (1.04)	4.372 (1.51)	3.871* (1.79)	1.471 (0.93)	4.284 (1.49)
<i>PASS</i>	-2.906 (-0.46)	-10.099*** (-4.40)	-2.500 (-0.62)	-3.200 (-0.50)	-10.519*** (-4.51)	-2.813 (-0.68)
<i>INTTR</i>	6.660*** (2.71)	1.011 (1.47)	12.354*** (5.82)	6.635*** (2.71)	0.714 (0.75)	12.314*** (5.69)
<i>DIVER</i>	-2.822* (-1.87)	-0.347 (-1.15)	-1.146 (-1.468)	-2.855* (-1.89)	-0.390 (-1.22)	-1.177 (-1.51)
<i>FSIZE</i>	-0.092 (-0.26)	0.296*** (2.80)	0.513** (2.34)	-0.056 (-0.16)	0.301*** (2.73)	0.536*** (-2.06)
<i>IDUMMY</i>	Yes	Yes	Yes	Yes	Yes	Yes
<i>YDUMMY</i>	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R ²	0.191	0.708	0.364	0.192	0.696	0.364
No. of firms	420	434	854	420	434	854

The dependent variable is *EDIF*. The observed exchange rate exposures are estimated using the equally-weighted KOSPI market index or monthly stock returns for firms and the market index. *FWD-Buy* and *FWD-Sell* are buy- and sell-transaction amount of currency derivatives, respectively. *NSFWD* is net position in structured forward contracts. *NFCFIN* is net foreign currency financing amount. *PASS* is exchange rate pass-through ratio. *INTTR* is internal transactions with foreign subsidiaries. *DIVER* is diversification index. *FSIZE* is firm size. *IDUMMY* and *YDUMMY* are dummy variables for industry and year, respectively, whose results are not reported here for brevity's sake. t-statistics are in parentheses.

* denotes significance at the 10% level.

** denotes significance at the 5% level.

*** denotes significance at the 1% level.

Appendix

Korea Standard Industrial Classification (KSIC) Code and Industry

KSIC Code	Definition of Industry
01	Agriculture
02	Forestry
03	Fishing
05	Mining of Coal, Crude Petroleum and Natural Gas
06	Mining of Metal Ores
07	Mining of Non-metallic Minerals, Except Fuel
08	Mining Support Service Activity
10	Manufacture of Food Products
11	Manufacture of Beverages
12	Manufacture of Tobacco Products
13	Manufacture of Textiles, Except Apparel
14	Manufacture of Wearing Apparel, Clothing Accessories and Fur Articles
15	Tanning and Dressing of Leather, Manufacture of Luggage and Footwear
16	Manufacture of Wood and of Products of Wood and Cork, Except Furniture
17	Manufacture of Pulp, Paper and Paper Products
18	Printing and Reproduction of Recorded Media
19	Manufacture of Coke, Hard-Coal and Lignite Fuel Briquettes and Refined Petroleum Products
20	Manufacture of Chemicals and Chemical Products Except Pharmaceuticals, Medicinal Chemicals
21	Manufacture of Pharmaceuticals, Medicinal Chemicals and Botanical Products
22	Manufacture of Rubber and Plastic Products
23	Manufacture of Other Non-metallic Mineral Products
24	Manufacture of Basic Metal Products
25	Manufacture of Fabricated Metal Products, Except Machinery and Furniture
26	Manufacture of Electronic Components, Radio, Television and Communication Equipment and Apparatuses
27	Manufacture of Medical, Precision and Optical Instruments, Watches and Clocks
28	Manufacture of Electrical Equipment
29	Manufacture of Other Machinery and Equipment
30	Manufacture of Motor Vehicles, Trailers and Semitrailers
31	Manufacture of Other Transport Equipment
32	Manufacture of Furniture
33	Other Manufacturing
35	Electricity, Gas, Steam and Air Conditioning Supply
36	Water Supply
37	Sewage, Wastewater and Human Waste Treatment Services
38	Waste Collection, Disposal and Materials Recovery
39	Remediation Activities and Other Waste Management Services
41	General Construction
42	Special Trade Construction

45	Sale of Motor Vehicles and Parts
46	Wholesale Trade and Commission Trade, Except of Motor Vehicles and Motorcycles
47	Retail Trade, Except Motor Vehicles and Motorcycles
49	Land Transport ; Transport Via Pipelines
50	Water Transport
51	Air Transport
52	Storage and Support Activities for Transportation
55	Accommodation
56	Food and Beverage Service Activities
58	Publishing Activities
59	Motion Picture, Video and Television Program Production, Sound Recording and Music Publishing Activities
60	Broadcasting
61	Telecommunications
62	Computer Programming, Consultancy and Related Activities
63*	Information Service Activities
64*	Financial Institutions, Except Insurance and Pension Funding
65*	Insurance and Pension Funding
66*	Activities Auxiliary to Financial Service and Insurance Activities
68	Real Estate Activities
69	Renting and leasing, Except Real Estate
70	Research and Development
71	Professional Services
72	Architectural, Engineering and Other Scientific Technical Services
73	Professional, Scientific and Technical Services
74	Business Facilities Management and Landscape Services
75	Business Support Services
84	Public Administration and Defense ; Compulsory Social Security
85	Education
86	Human Health
87	Social Work Activities
90	Creative, Arts and Recreation Related Services
91	Sports Activities and Amusement Activities
94	Membership Organizations
95	Maintenance and Repair Services
96	Other Personal Services Activities
97	Private Households with Employed Persons
98	Undifferentiated Goods- and Services- Producing Activities of Private Households
99	Extra-Territorial Organization and Bodies

* denotes KSIC codes of financial firms which are excluded for analysis in our study.